Can Social Networks Predict Running Behavior?

Social Aggregation and Graph Neural Networks on Strava

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Outline

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Section 1

Introduction

Motivation and research question

Context

- Strava: one of the most used sport platforms and social networks among athletes
- Prior work: mostly psychological (self-presentation, social pressure, motivation) or SIENA-based influence models (Franken et al., 2022a)

This study (same data as Franken et al., 2022a; no influence estimation)

1. To what extent does Strava work as an **infrastructure for social aggregation**?
2. Are **high-status athletes** more embedded or less bridging than other athletes, once the number of ties is taken into account?
3. Can **contacts improve the prediction** of running behavior beyond the athlete's own history?

Diffusion from high-status athletes is only partially addressed: the models do not test contact status directly.

Theory: weak ties carry behavior across groups (Granovetter, 1973; Burt, 2004); dense groups reinforce it through repeated exposure (Centola, 2010).

Section 2

Data and Methods

Data and analysis pipeline

Data: reproduction package of Franken et al. (2022a), RSiena format

- 5 Strava running clubs, **329 athletes**, 12 monthly waves in 2019 (3,948 athlete-month obs.)
- Running frequency and volume, other sport activity, **kudos** exchanged within the club

Club	1	2	3	4	5	Total
Athletes	27	58	159	9	76	329

Notebook structure

- **Part I** — loading, panel/edge tables, exploratory analysis
- **Part II** — network analysis, homophily, brokerage and embeddedness
- **Part III** — leakage-free prediction: XGBoost and GraphSAGE

Clubs 1 and 4 are small: reported, but given limited weight.

Exploratory findings

- Behavior is **very stable**: in 57% of athlete-month transitions the weekly category does not change
- Stable between-athlete differences explain about **74%** of the variance in running frequency
- Mean kudos given/received: from 2.0 (Club 3) to 11.1 (Club 2); right-skewed in all clubs
- Running volume in Club 5 is on a wider scale (0–9): not directly comparable
- Strava tenure is **not** a measure of running experience

Implication

Little month-to-month change: a person's own history is already a strong predictor, a hard baseline for any network-based model.

Network construction

Graph	Direction	Time	Tie	Used for
Kudos	directed, weighted	year	A gave ≥ 1 kudos to B (weight = n. months)	status (in-degree), ego networks, degree distribution
Reciprocal	undirected, weighted	year	A and B exchanged kudos in the **same month**	strong ties: bridges, embeddedness
Monthly snapshots	symmetrised	month t	kudos given in month t	GNN input (information up to t)

- **Status**: in-degree in the annual kudos graph above the club median
- **Local bridge**: strong tie whose ends share no contact
- **Overlap**: common contacts / distinct contacts of the two ends, in $[0, 1]$ (high = tie inside a dense group)

Section 3

Results: Network Analysis

Strava as a social aggregation structure

	C1	C2	C3	C4	C5
Athletes	27	58	159	9	76
Density	0.165	0.304	0.036	0.403	0.190
Months per tie	7.4	7.5	3.9	7.4	6.3
Reciprocity	0.845	0.816	0.556	0.828	0.826
Transitivity (strong ties)	0.61	0.54	0.36	0.72	0.54
Largest component	0.59	0.93	0.52	0.89	0.82
Mean path length	1.7	1.8	3.3	2.0	2.0
Assortativity (commitment)	0.01	0.00	0.05	-0.02	0.09

- Kudos form **cohesive, reciprocal communities** in four clubs: reciprocity 0.82–0.85, strong-tie transitivity 0.54–0.72, paths of 1.7–2.0 steps
- Mean kudos degree: about 17 (Club 2) and 14 (Club 5), against 3–4 in Club 3; density falls with club size (0.036 in Club 3)
- **Club 3** (largest) is the exception: only about 50% of ties are mutual, ties last 3.9 months (6.3–7.5 elsewhere), 15 athletes with no kudos tie and 74 of 159 (47%) without a strong tie; mean path 3.3, diameter 8
- Athletes do **not** sort by commitment level

Permutation tests: design

Question: do high- and low-status (or high- and low-commitment) athletes differ in bridging or embeddedness? Reciprocal graph, strong ties only.

$$D_{obs} = \frac{1}{|H|}\sum_{i\in H} y_i - \frac{1}{|L|}\sum_{i\in L} y_i, \qquad p = \frac{1 + \sum_{b=1}^{B} \mathbb{1}\{|D^{(b)}| \geq |D_{obs}|\}}{B+1}, \quad B = 5000$$

- y_i: share of bridging ties (local bridges) or mean overlap of own ties (embeddedness)
- Bridges: labels shuffled **within club**; robustness check excluding athletes with a single tie and shuffling **within club and degree**
- Embeddedness: athletes with a single tie left out; labels shuffled **within club and degree**, so that the number of ties cannot explain a difference

An athlete with a single tie has a bridging share of 1 by construction.

Status, bridges and embeddedness: results

Outcome (shuffling)	Grouping	D	p
Bridging share, ≥ 1 tie (club)	Status	-0.232	< 0.001
Bridging share, ≥ 1 tie (club)	Commitment	-0.024	0.609
Bridging share, ≥ 2 ties (club, degree)	Status	$+0.034$	0.631
Bridging share, ≥ 2 ties (club, degree)	Commitment	$+0.046$	0.126
Mean overlap, ≥ 2 ties (club, degree)	Status	$+0.058$	0.734
Mean overlap, ≥ 2 ties (club, degree)	Commitment	$+0.039$	0.875

- The raw gap in bridging (-0.23) comes from athletes with a single tie: **27 of the 29 are low-status**, and their only tie is a bridge by construction
- With the number of ties held fixed ($n = 202$), high-status athletes are **neither less bridging nor more embedded**
- Limited power: only 62 low-status athletes with at least two ties (7 of them in Club 3)

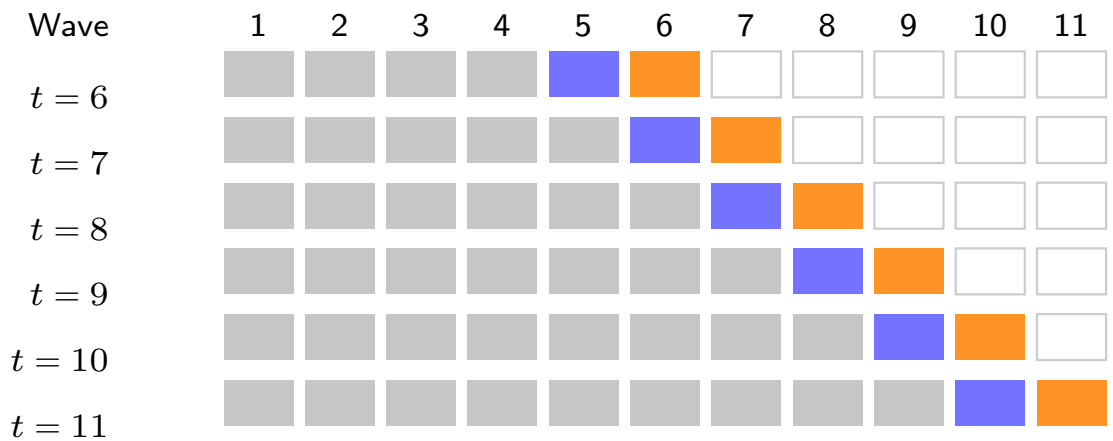
Section 4

Results: Prediction

Leakage-free predictive design

Target: does weekly running frequency **increase** from month t to $t+1$? (eligible athletes only: in the club in both months, not at the scale ceiling)

- All features use information **up to** t (cumulative commitment and status)
- Automatic **truncation test**: rebuilding features as if data ended earlier must not change past values
- Preprocessing (log, standardisation) fitted on training rows only



gray: training; blue: last training transition (tunes trees/epochs); orange: test month; white: future, never used

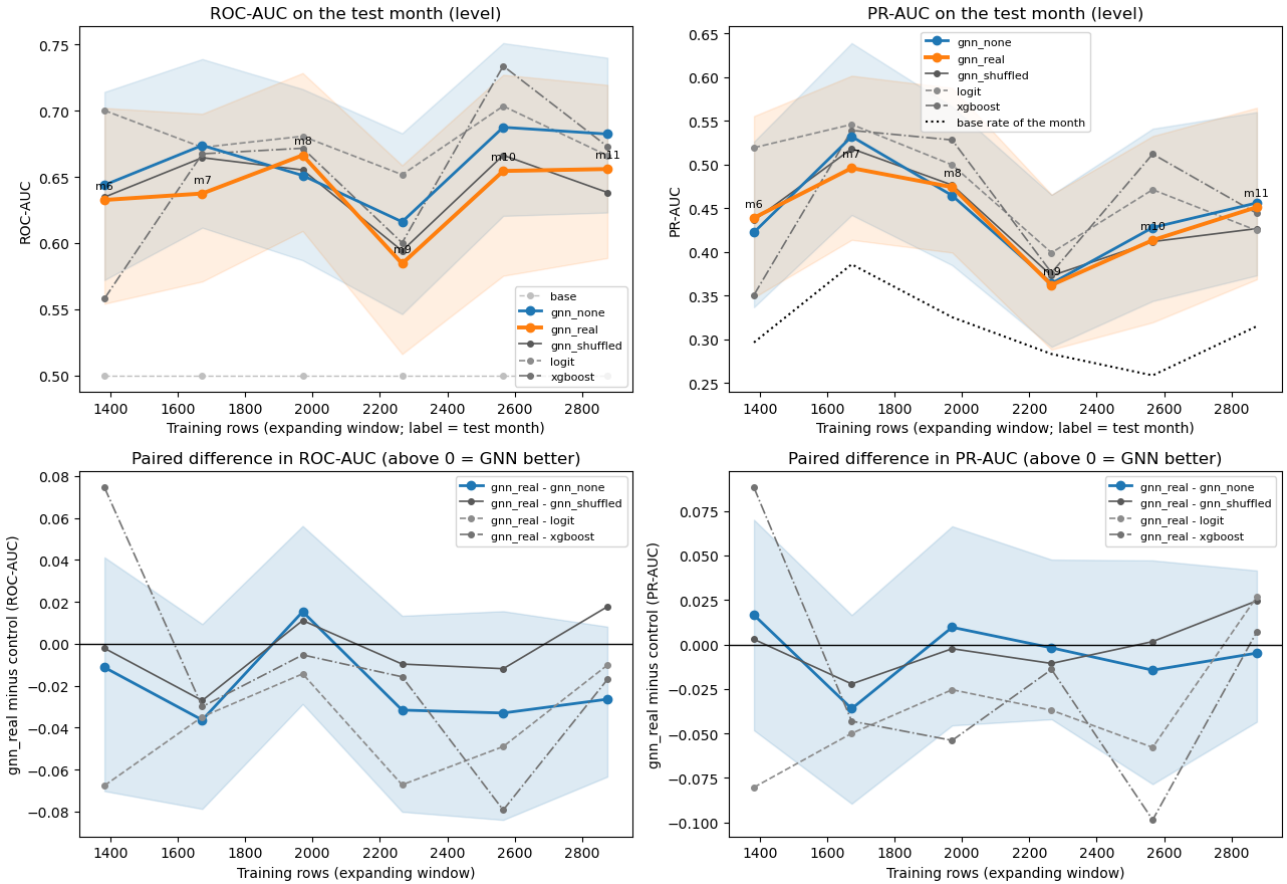
Models compared

Model	Input	Contacts?
Base rate	training outcomes only	no
Logistic regression	own features at t	no
XGBoost	own features, **contact summaries**, club, calendar month	summaries
GraphSAGE (2 layers)	club snapshot graph at t	**real edges**
GraphSAGE, no edges	same, edges removed	control
GraphSAGE, rewired	same, randomly rewired edges	control

$$h_i^{(l)} = \mathrm{ReLU}\Big(W_1^{(l)} h_i^{(l-1)} + W_2^{(l)} \,\mathrm{mean}_{j \in N_t(i)} h_j^{(l-1)}\Big), \quad \hat{p}_i = \sigma(w^\top h_i^{(2)} + b)$$

- Evaluation: ROC-AUC and PR-AUC on **six expanding-window folds** (test months 6–11); paired bootstrap on athletes
- No-edge and rewired controls isolate the contribution of the **real contacts**

Out-of-time performance



	Base	GNN-none	GNN-real	GNN-rew.	Logit	XGB
Mean ROC-AUC (months 6–11)	0.500	0.659	0.638	0.642	**0.679**	0.651

ROC-AUC by test month

Month	Base	GNN-none	GNN-real	GNN-rew.	Logit	XGB
6	0.500	0.644	0.633	0.635	0.700	0.558
7	0.500	0.674	0.637	0.664	0.672	0.667
8	0.500	0.651	0.666	0.655	0.681	0.672
9	0.500	0.616	0.584	0.594	0.652	0.600
10	0.500	0.687	0.654	0.666	0.703	0.734
11	0.500	0.682	0.656	0.638	0.666	0.673
Mean	0.500	0.659	0.638	0.642	**0.679**	0.651

Each test month has about 300 observations (single-month ROC s.e. about 0.04): month-by-month differences should not be over-interpreted.

Do contacts add information?

- All models beat the base rate: next-month increases are **partly predictable** (top-decile lift about 1.5–1.7)
- **Real edges do not help**: GNN real vs. no edges, ROC difference -0.018 (95% CI $[-0.035, -0.001]$), PR-AUC difference $+0.006$ (CI $[-0.015, +0.028]$); vs. rewired, ROC -0.002 (CI $[-0.016, +0.011]$)
- A simple model on the athlete's own history is as good as, or better than, any model using the network: logistic regression beats the real-edge GNN (ROC difference -0.036, CI $[-0.055, -0.018]$)
- XGBoost with explicit exposure variables does not improve on logistic regression
- Caution: about 300 observations per test month (single-month ROC s.e. about 0.04)

Take-away

Contacts at month t add nothing beyond the athlete's own recent behavior. This is *consistent with*, but does not prove, the absence of social influence: the data are small and observational, the outcome is a coarse ordinal category, and influence may act over longer lags or through unrecorded channels.

Section 5

Conclusions

Limitations and conclusions

Limitations

- Few clubs, two with fewer than 30 athletes; about 300 test observations per month
- Kudos are a proxy for ties; tenure is not running experience; ordinal outcome with a ceiling
- Observational data: influence cannot be separated from selection into ties
- Monthly resolution and one-month horizon may miss slower diffusion; annual aggregation hides tie timing
- Low-status group is small (limited power of the permutation tests)
- Predictive models do not test the role of **contact status** directly

Conclusions

1. Kudos form cohesive, reciprocal communities (except the largest club)
2. Once the number of ties is taken into account, high-status athletes are **neither more embedded nor less bridging**: the raw bridging gap is explained by the many low-status athletes with a single tie
3. Contacts do not improve next-month prediction beyond the athlete's own recent history: a **limit** of social influence in this setting, not evidence of its absence

Possible extensions: longer lags, status-weighted exposure, directed edges, larger samples

Thank you

Questions?

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